

Big Picture: From Thermodynamic Laws to Real Fluid Calculations

CBE 20260 / Thermodynamics I

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Table of contents

1	Why This Section Exists	2
1.1	The Problem	2
2	The Big Idea	3
2.1	The Goal	3
3	Step 1: Equations of State (EOS)	3
3.1	Key Insight	4
4	Step 2: The Fundamental Equation	4
4.1	Big Interpretation	4
5	Step 3: Changing Variables (Legendre Transforms)	5
5.1	Result: Four Potentials	5
6	Step 4: Maxwell Relations (The Key Tool)	5
7	Step 5: General Property Equations	6
7.1	Enthalpy	6
7.2	Internal Energy	6
7.3	Big Result	6
8	Step 6: The Roadblock	6
9	Step 7: The Breakthrough — Residual Properties	7
9.1	Why This Works	7
9.2	Master Equations	7
9.3	Big Insight	7
10	Step 8: Equations of State Become Powerful	8
10.1	Examples	8

11 Step 9: Generalized Correlations	8
11.1 Interpretation	9
12 Step 10: Final Toolbox	9
13 The Big Picture (One Slide Version)	9

1 Why This Section Exists

At the beginning of the course, we solved problems like:

- Turbines, compressors, heat exchangers
- Rankine cycles
- Refrigeration cycles

using:

- Ideal gas relations
- Steam tables

These worked well—but only for **specific fluids and tabulated data**.

1.1 The Problem

- What if: - The fluid is new (no tables exist)?
- The pressure is very high?
- The fluid is not water?

You cannot rely on tables.

You need a **general method**.

2 The Big Idea

The entire section is built around one fundamental postulate:

A simple compressible system is completely defined by two independent intensive properties

So if we know:

$$P, V, T$$

then in principle we can determine:

$$U, H, S, G$$

2.1 The Goal

Build a framework to compute ALL thermodynamic properties from: - an equation of state (EOS)
- measurable variables (P, V, T)

3 Step 1: Equations of State (EOS)

We began by introducing equations of state:

- Ideal gas:

$$PV = RT$$

- Real fluids:

- van der Waals
- Peng–Robinson
- SRK

These relate:

$$P = f(T, V)$$

3.1 Key Insight

An EOS gives you **P–V–T behavior**, but engineers need:

- enthalpy
- entropy
- internal energy

□ So we need a way to connect **PVT → all properties**

4 Step 2: The Fundamental Equation

From the First + Second Laws:

$$dU = TdS - PdV$$

This is the **starting point of everything**.

It tells us:

- $U = U(S, V)$
 - $T = \left(\frac{\partial U}{\partial S}\right)_V$
 - $P = -\left(\frac{\partial U}{\partial V}\right)_S$
-

4.1 Big Interpretation

Temperature and pressure are **slopes of a thermodynamic surface**
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5 Step 3: Changing Variables (Legendre Transforms)

We don't want to work in (S, V) .

We want:

- T and P (measurable!)

So we transform:

- $U \rightarrow H = U + PV$
- $U \rightarrow A = U - TS$
- $U \rightarrow G = U + PV - TS$

5.1 Result: Four Potentials

Each has its own "natural variables":

Potential	Natural Variables
U	(S, V)
H	(S, P)
A	(T, V)
G	(T, P)

6 Step 4: Maxwell Relations (The Key Tool)

From exact differentials:

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$$

These allow us to:

Replace **unmeasurable derivatives (entropy)** with **measurable ones (PVT)**

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7 Step 5: General Property Equations

Using calculus + Maxwell relations, we derived:

7.1 Enthalpy

$$dH = C_p dT + \left[V - T \left(\frac{\partial V}{\partial T} \right)_P \right] dP$$

7.2 Internal Energy

$$dU = C_v dT + \left[T \left(\frac{\partial P}{\partial T} \right)_V - P \right] dV$$

7.3 Big Result

If you know an EOS (i.e., $V = f(T, P)$), you can compute H , U , and S .

8 Step 6: The Roadblock

In real problems:

$$\Delta H = \int C_p(T, P) dT + \text{complicated term}$$

But:

- We only have **ideal gas** $C_p(T)$ **data**
- EOS integrals are messy

□ This is where students usually get lost.

9 Step 7: The Breakthrough — Residual Properties

Define:

$$M^R = M - M^{IG}$$

So:

$$M = M^{IG} + M^R$$

9.1 Why This Works

We split the problem:

1. Ideal gas part → easy (tables or Cp correlations)
 2. Residual part → from EOS
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9.2 Master Equations

$$\Delta H = \Delta H^{IG} + (H_2^R - H_1^R)$$

$$\Delta S = \Delta S^{IG} + (S_2^R - S_1^R)$$

9.3 Big Insight

EOS gives you the **correction to ideal gas behavior**

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10 Step 8: Equations of State Become Powerful

Now EOS is not just:

$$P = f(T, V)$$

It becomes a tool to compute:

- Z
 - H^R
 - S^R
 - U^R
-

10.1 Examples

- van der Waals
 - Peng–Robinson
 - SRK
-

11 Step 9: Generalized Correlations

Instead of solving EOS every time:

- Use reduced properties:

$$T_r = \frac{T}{T_c}, \quad P_r = \frac{P}{P_c}$$

- Use acentric factor ω

Then:

$$Z = Z^{(0)} + \omega Z^{(1)}$$

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11.1 Interpretation

Many fluids behave similarly when scaled properly

12 Step 10: Final Toolbox

At this point, you can solve:

- Compressors (real gases)
- Turbines
- Heat exchangers
- Phase equilibria

WITHOUT tables.

13 The Big Picture (One Slide Version)

**PVT Data + EOS → Thermodynamic Identities → Maxwell Relations
→ General Property Equations → Residual Properties → Real Engineering
Calculations**